

ELI HEADLEY

Atlanta, GA · 678.953.4496 · eliheadley@gmail.com
linkedin.com/in/eli-headley · github.com/eliheadley · orcid.org/0009-0008-6721-8342

SUMMARY

Computer Vision Engineer with production experience building, testing, and deploying deep learning systems at industrial scale. Experienced in edge inference, model serving, and distributed event-driven systems, with a background in published deep learning research for biometric recognition.

EDUCATION

Kennesaw State University

Bachelor of Science, Computer Science

GPA: 4.0 / 4.0

Relevant coursework: Computer Vision, Machine Learning, Linear Algebra, Algorithm Analysis, Parallel & Distributed Computing, Operating Systems, Data Structures, Computer Organization & Architecture.

Marietta, GA

Dec. 2024

PUBLICATIONS

Headley, E., Tebor, D., & Karakaya, M (2024). How Eye Blink Affects the Recognition Performance of Traditional Off-Angle Iris Images. SoutheastCon 2024, Atlanta, GA, pp. 1243–1249. IEEE.

Tebor, D., Headley, E., & Karakaya, M. (2024). Impact of Blinking on Deep Learning Based Iris Recognition. In Proceedings of the 2024 ACM Southeast Conference (ACMSE '24), Marietta, GA, pp. ACM.

EXPERIENCE

Georgia-Pacific

Computer Vision Engineer

Atlanta, GA

Jan. 2025 – Present

- Helped grow a manufacturing safety application from first deployment to 70+ manufacturing sites and 500+ cameras in production.
- Hardened the pipeline and built monitoring and alerting that sustained 99% uptime across that fleet.
- Deployed production computer vision models on NVIDIA Triton Inference Server and DeepStream pipelines, partnering with an internal data science team that owns model training.
- Designed and maintained containerized Python services running on Kubernetes.
- Built event-driven integrations using NATS to move detection events between edge inference and downstream consumers.

Kennesaw State University, College of Computing and Software Engineering

Undergraduate Research Assistant

Marietta, GA

Feb. 2023 – May 2024

- Ran occlusion experiments on frontal and off-angle iris images from ~100 subjects, measuring how eyelid occlusion from blinking affects recognition in both traditional Gabor-based and CNN-based (ResNet) pipelines.
- Established that training on occluded images largely offsets the accuracy loss, and that partial occlusion can improve performance on severe off-angle images by masking distortion-prone regions.
- Co-authored both resulting papers on a three-person research team.

Kennesaw State University Housing and Residence Life

Resident Assistant

Marietta, GA

Aug. 2022 – May 2024

- Supported a community of 35 residents; planned community programming and resolved day-to-day conflicts.

PROJECTS

Entrio — Distributed Computer Vision Pipeline

Mar. 2026 – Present

- Edge inference application running YOLOv11 through NVIDIA DeepStream on a Jetson Orin Nano, with a multithreaded Python pipeline handling alert logic, recording, and publishing.
- Go API service backed by a PostgreSQL database with schema migrations managed by Flyway, plus a React Native front end for the UI.

Decaf Interpreter — Bytecode Virtual Machine

Jan. 2026 – Present

- Implementing a bytecode virtual machine and interpreter for the Decaf language in C, covering scanning, parsing, instruction dispatch, and memory management.

TECHNICAL SKILLS

Languages: Python, Go, C, Java, SQL, Terraform

ML & Computer Vision: NVIDIA DeepStream, Triton Inference Server, YOLOv11, TensorFlow Lite, OpenCV

Infrastructure & Data: Docker, Kubernetes, AWS (S3, IAM, Lambda, ECS, CloudWatch), NATS, PostgreSQL, Neo4j

Hardware & Edge: NVIDIA Jetson Orin Nano, JetPack, RTSP